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### Image device and motherboard for image device

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#### Abstract

At a motherboard for an image device, which is used for manufacturing an image device such as a liquid crystal device, an organic electroluminescence device, a mirror device, and an image-capturing device, a mark for alignment is provided on an outer side of a pixel area in which a plurality of pixels are arranged. In the photo-lithography process, a light exposure mask is arranged by using the mark as a reference position. For the mark, a recessed portion provided in a motherboard main body is filled with a filling film. The recessed portion includes a first groove extending along a first direction and a second groove extending along a second direction intersecting with the first direction, the second groove not intersecting with the first groove in plan view.

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## Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2021-077279, filed Apr. 30, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

### BACKGROUND

#### 1. Technical Field

(2) The present disclosure relates to an image device that enables displaying and imaging and a motherboard for an image device.

#### 2. Related Art

(3) An image device such as a liquid crystal device, a display device including a light-emitting element such as an organic electroluminescence element, a mirror device, and an image sensor

obtained by combining a photoreceptor element and a lens performs displaying and imaging with a plurality of pixels provided in a pixel area. In a process of manufacturing the image device, a semiconductor process such as a photo-lithography process is utilized to form various elements, wiring lines, and the like on a substrate body. Further, in order to expose a resist to light in the photo-lithography process, a mark is formed at the substrate body, and a light exposure mask is positioned by utilizing the mark, for example. The mark has a structure obtained by filling, with a filling film, a quadrangular or cross-like recessed portion formed in the substrate body, for example (see JP-A-2013-89881).

(4) In a case in which the mark is formed, when the recessed portion is formed to have a quadrangular shape, an opening width is large, and hence there is a difficulty in filling the recessed portion with the filling film in an appropriate manner. Further, when the recessed portion is formed to have a cross-like shape, an opening width in an oblique direction with respect to both a first groove and a second groove is large at a crossing portion of the first groove extending in a first direction and the second groove extending in a second direction, and hence there is a difficulty in filling the recessed portion with the filling film in an appropriate manner. In such a case, there arises a problem in that a crack or the like generated in the recessed portion due to stress may make it difficult to recognize the mark.

## SUMMARY

(5) In order to solve the above-mentioned problem, an image device according to an aspect of the present disclosure includes a substrate body and a mark including a recessed portion provided in the substrate body and a filling film filled in the recessed portion. The recessed portion includes a first groove extending along a first direction and a second groove extending along a second direction intersecting with the first direction, the second groove not intersecting with the first groove in plan view.

(6) A motherboard for an image device according to an aspect of the present disclosure includes a motherboard main body and a mark including a recessed portion provided in the motherboard main body and a filling film filled in the recessed portion. The recessed portion includes a first groove extending along a first direction in an in-plane direction of the motherboard main body and a second groove extending along a second direction intersecting with the first direction, the second groove not intersecting with the first groove in plan view.

(7) In another aspect of the present disclosure, a method of manufacturing an image device includes forming a mark as a reference, the mark including a recessed portion recessed on an outer side of a pixel area in which a plurality of pixels are arranged. In the method, the recessed portion including a first groove and a second groove is formed. The first groove extends in a first direction in an in-plane direction of a substrate body, and the second groove extends in a second direction intersecting with the first groove, the second groove not intersecting with the first groove in the in-plane direction. After forming the recessed portion, a filling film fills the recessed portion.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view illustrating an aspect of an image device according to a first exemplary embodiment of the present disclosure.

(2) FIG. 2 is a cross-sectional view of the image device illustrated in FIG. 1.

(3) FIG. 3 is a plan view of a plurality of pixels adjacent to each other in the image device illustrated in FIG. 1.

(4) FIG. 4 is a cross-sectional view of a vicinity of a transistor illustrated in FIG. 3, which is taken along the line G-G'.

(5) FIG. 5 is a cross-sectional view of the vicinity of the transistor illustrated in FIG. 3, which is

taken along the line S-S'.

(6) FIG. 6 is an explanatory view of a motherboard for an image device according to the first exemplary embodiment of the present disclosure.

(7) FIG. 7 is a plan view illustrating a mark illustrated in FIG. 1 in an enlarged manner.

(8) FIG. 8 is a cross-sectional view of the mark illustrated in FIG. 7.

(9) FIG. 9 is a plan view illustrating a mark according to a reference example of the present disclosure in an enlarged manner.

(10) FIG. 10 is a cross-sectional view of the mark illustrated in FIG. 9.

(11) FIG. 11 is an explanatory view illustrating a second configuration example of the mark at the motherboard for an image device according to the present disclosure.

(12) FIG. 12 is an explanatory view illustrating a third configuration example of the mark at the motherboard for an image device according to the present disclosure.

(13) FIG. 13 is an explanatory view illustrating a fourth configuration example of the mark at the motherboard for an image device according to the present disclosure.

(14) FIG. 14 is an explanatory view illustrating a fifth configuration example of the mark at the motherboard for an image device according to the present disclosure.

(15) FIG. 15 is an explanatory view illustrating a sixth configuration example of the mark at the motherboard for an image device according to the present disclosure.

(16) FIG. 16 is an explanatory view illustrating a seventh configuration example of the mark at the motherboard for an image device according to the present disclosure.

(17) FIG. 17 is a plan view illustrating a configuration example of a capacitance element illustrated in FIG. 4 and FIG. 5.

(18) FIG. 18 is an explanatory view of a modification example of the motherboard for an image device illustrated in FIG. 6.

(19) FIG. 19 is an explanatory view of an organic electroluminescence display device to which the present disclosure is applied.

(20) FIG. 20 is an explanatory view of a mirror device to which the present disclosure is applied.

(21) FIG. 21 is an explanatory view of an image-capturing device to which the present disclosure is applied.

## DESCRIPTION OF EXEMPLARY EMBODIMENTS

(22) Exemplary embodiments of the present disclosure will be described below with reference to the drawings. An “image device” according to the present disclosure means a display device that displays an image or an image-capturing device that captures an image. Therefore, in the following description, an application example to a liquid crystal device **1a**, an application example to an organic electroluminescence device **1b**, an application example to a mirror device **1c**, and an application example to an image-capturing device **1d** are described in a first exemplary embodiment, a second exemplary embodiment, a third exemplary embodiment, and a fourth exemplary embodiment, respectively. Note that, in each of the figures to be referred to in the following description, to illustrate each layer, each member, and the like in a recognizable size in the drawings, each layer, each member, and the like are illustrated at a different scale. Further, in the following description, a first direction of two directions intersecting with each other in an in-plane direction of an element substrate **10** is denoted with X, and a second direction intersecting with the first direction X is denoted with Y.

### 1. First Exemplary Embodiment: Application Example to Liquid Crystal Device **1a**

#### (23) 1-1. Configuration of Image Device **1**

(24) FIG. 1 is a plan view illustrating an aspect of an image device **1** according to the first exemplary embodiment of the present disclosure. FIG. 2 is a cross-sectional view of the image device **1** illustrated in FIG. 1. The image device **1** illustrated in FIG. 1 and FIG. 2 corresponds to the liquid crystal device **1a**. In the liquid crystal device **1a**, the element substrate **10** including a substrate body **19** and a counter substrate **20** including a substrate body **29** are bonded to each other

through intermediation of a sealing material **107** at a predetermined gap, and an electro-optical layer **80** is arranged in a region surrounded by the sealing material **107** between the element substrate **10** and the counter substrate **20**. A pixel area **10b** is provided to the substantially center part of the image device **1**.

(25) The substrate body **19** is formed of a quartz substrate, a glass substrate, or the like. At one surface **19s** side of the substrate body **19**, on an outer side of the pixel area **10b**, a data line driving circuit **101** and a plurality of terminals **102** are formed along one side of the substrate body **19**. A scanning line driving circuit **104** is formed along other sides adjacent to the one side. A flexible wiring substrate (not illustrated) is coupled to the terminals **102**.

(26) In the pixel area **10b** on the one surface **19s** of the substrate body **19**, pixels **1e** including a plurality of pixel electrodes **9a** being transmissive and being formed of an Indium Tin Oxide (ITO) film or the like and a transistor (not illustrated in FIG. 1 and FIG. 2) electrically coupled to each of the plurality of pixel electrodes **9a** are formed in a matrix shape. A first alignment film **18** is formed at the counter substrate **20** side with respect to the pixel electrodes **9a**. Therefore, the element substrate **10** corresponds to a part from the substrate body **19** to the first alignment film **18**. At a region on an outer side of the pixel area **10b** of the substrate body **19**, marks **M** described later are formed.

(27) The substrate body **29** is formed of a quartz substrate, a glass substrate, or the like. At one surface **29s** side of the substrate body **29**, a light-transmitting common electrode **21** being formed of an ITO film or the like is formed. The common electrode **21** is formed substantially entirely at the counter substrate **20**. A second alignment film **28** is formed at the element substrate **10** side with respect to the common electrode **21**. Therefore, the counter substrate **20** corresponds to a part from the substrate body **29** to the second alignment film **28**. At the counter substrate **20**, a light-shielding layer **27** having a light-shielding property and being formed of a metal or a metal compound is formed between the substrate body **29** and the common electrode **21**, and a protective layer **26** being transmissive is formed between the light-shielding layer **27** and the common electrode **21**. The light-shielding layer **27** is formed, for example, as a partition **27a** in a frame-like shape extending along the outer peripheral edge of the pixel area **10b**. In some cases, the light-shielding layer **27** is formed as a black matrix **27b** in regions that overlap with regions in plan view, each of which is sandwiched between the pixel electrodes **9a** adjacent to each other. Regions overlapping, when viewed in plan view, with the partition **27a** at the substrate body **19** are formed with dummy pixel electrodes **9d** formed simultaneously with the pixel electrodes **9a**.

(28) The first alignment film **18** and the second alignment film **28** are inorganic alignment films including diagonally vapor-deposited films of  $\text{SiO.sub.x}$  ( $x < 2$ ),  $\text{SiO.sub.2}$ ,  $\text{TiO.sub.2}$ ,  $\text{MgO}$ ,  $\text{Al.sub.2O.sub.3}$ , and the like, where liquid crystal molecules having negative dielectric anisotropy used for the electro-optical layer **80** are tilt-oriented. Therefore, the liquid crystal molecules form a predetermined angle with respect to the substrate body **19** and the substrate body **29**. In this way, the image device **1** is configured as a liquid crystal device of a Vertical Alignment (VA) mode.

(29) At the element substrate **10**, inter-substrate conduction electrodes **109** for establishing electrical conduction between the element substrate **10** and the counter substrate **20** are formed in regions overlapping with corner portions of the substrate body **29** on the outer side of the sealing material **107**. An inter-substrate conduction material **109a** containing conductive particles is arranged at the inter-substrate conduction electrode **109**, where the common electrode **21** of the counter substrate **20** is electrically coupled to the element substrate **10** side via the inter-substrate conduction material **109a** and the inter-substrate conduction electrode **109**. Hence, a common potential is applied to the common electrode **21** from the element substrate **10** side.

(30) In the image device **1** of the present exemplary embodiment, the pixel electrodes **9a** and the common electrode **21** are formed of an ITO film, and the image device **1** is configured as a transmissive liquid crystal device. In the image device **1** described herein, an image is displayed in a manner that light incident on the electro-optical layer **80** from one of the element substrate **10** and

the counter substrate **20** is modulated while being emitted through the other one of the substrates. In the present exemplary embodiment, as indicated by the arrow **L**, an image is displayed in a manner that light incident from the counter substrate **20** is modulated by the electro-optical layer **80** for each of the pixels while being emitted through the element substrate **10**.

(31) 1-2. Overall Configuration of Pixel **1e**

(32) FIG. **3** is a plan view of the plurality of pixels **1e** adjacent to each other in the image device **1** illustrated in FIG. **1**. FIG. **4** is a cross-sectional view of a vicinity of a transistor **30** illustrated in FIG. **3**, which is taken along the line G-G'. FIG. **4** schematically illustrates a state in which the transistor **30** and the like are taken along a scan line **3a**. Note that FIG. **4** also illustrates a cross section at a position passing through a contact hole **46e** and the like. FIG. **5** is a cross-sectional view of the vicinity of the transistor **30** illustrated in FIG. **3**, which is taken along the line S-S'. FIG. **5** schematically illustrates a state in which the transistor **30** and the like are taken along a data line **6a**. Note that, in FIG. **3**, each layer is indicated by a line described below. Further, in FIG. **3**, as for layers including ends overlapping with each other in plan view, positions of the ends are displaced to make shapes and the like of the layers readily recognizable. Scan line **3a**: a thick solid line Semiconductor film **31a**: a thin and short dashed line Gate electrode **33a**: a thin two-dot chain line First capacitance electrode **4a**: a thin dot chain line Second capacitance electrode **5a**: a thin solid line Relay electrodes **8c**, **8d**, **8s**: a thick dot chain line Data line **6a** and relay electrodes **6c** and **6e**: a thick two-dot chain line Capacitance line **7a** and relay electrode **7e**: a thick and long dashed line Pixel electrode **9a**: a thicker solid line

(33) As illustrated in FIG. **3**, the pixel electrode **9a** is formed in each of the plurality of pixels **1e** at the element substrate **10**, where the scan line **3a**, the data line **6a**, and the capacitance line **7a** extend along the inter-pixel region between the pixel electrodes **9a** adjacent to each other. More specifically, the scan line **3a** extends in the first direction X while overlapping with a first inter-pixel region **9b** extending in the first direction X, and the data line **6a** and the capacitance line **7a** extend in the second direction Y while overlapping with a second inter-pixel region **9c** extending in the second direction Y. At the element substrate **10**, the transistor **30** is formed corresponding to the intersection between the data line **6a** and the scan line **3a**. The scan line **3a**, the data line **6a**, and the capacitance line **7a** have a light-shielding property. Therefore, a region in which the scan line **3a**, the data line **6a**, the capacitance line **7a**, and a conductive film in the same layer as those lines are formed is a light-shielding region through which light is prevented from transmitting, and a region surrounded by the light-shielding region is an aperture region through which light transmits.

(34) As illustrated in FIG. **4** and FIG. **5**, in the element substrate **10**, insulating films **41**, **42**, **43**, **44**, **45**, and **46** as inter-layer insulating films are layered between the substrate body **19** and the pixel electrode **9a**. The insulating films **41**, **42**, **43**, **44**, **45**, and **46** are formed of a film being transmissive such as a silicon oxide film. Further, between the substrate body **19** and the insulating film **41**, there is provided a multi-layered film **550** in which the first conductive film **4a**, an insulating film **49** as a dielectric film, and a second conductive film **5a** are layered sequentially from the substrate body **19** side, and the multi-layered film **550** forms a capacitance element **55**. The insulating film **49** is formed of a film being transmissive such as a silicon nitride film and a silicon oxide film. When the insulating film **49** is formed of a silicon nitride film, the insulating film **49** is formed to have a film thickness from 0.01  $\mu\text{m}$  to 0.03  $\mu\text{m}$ . The first conductive film **4a** is formed of a conductive silicon film, a metal silicide film, a metal film, a metal compound film, or the like. When the first conductive film **4a** is formed of a conductive silicon film, the first conductive film **4a** is formed to have a film thickness from 0.03  $\mu\text{m}$  to 0.2  $\mu\text{m}$ . The second conductive film **5a** is formed of a conductive silicon film, a metal silicide film, a metal film, a metal compound film, or the like. When the second conductive film **5a** is formed of a conductive silicon film, the second conductive film **5a** is formed to have a film thickness from 0.03  $\mu\text{m}$  to 0.2  $\mu\text{m}$ .

(35) A recessed portion **190** recessed toward the substrate body **19** is formed between the substrate body **19** and the capacitance element **55**, and the depth of the recessed portion **190** is deeper than

the total film thickness of the multi-layered film **550**. Therefore, the second conductive film **5a** overlaps with a side wall and a bottom wall of the recessed portion **190** via the insulating film **49** and the first conductive film **4a**. Thus, the capacitance element **55** has electrostatic capacitance larger than that in a case in which the multi-layered film **550** is formed only on a flat surface. In the present exemplary embodiment, the recessed portion **190** is recessed in a surface of the substrate body **19** on the pixel electrode **9a** side, toward the substrate body **19** side.

(36) The scan line **3a** is formed at a layer between the insulating film **41** and the insulating film **42**. The scan line **3a** is a conductive light-shielding film **3b** formed of a metal silicide film, a metal film, a metal compound film, or the like. In the present exemplary embodiment, the scan line **3a** is formed of a tungsten silicide film.

(37) Between the insulating film **42** and the insulating film **43**, the transistor **30** for pixel switching is formed. The transistor **30** includes the semiconductor film **31a** formed at the surface of the insulating film **42** on the pixel electrode **9a** side, a gate insulating film **32** covering the semiconductor film **31a** from the pixel electrode **9a** side, and the gate electrode **33a** overlapping with a semiconductor film **31a** via the gate insulating film **32** in plan view.

(38) The gate electrode **33a** is formed of a multi-layered film including a conductive polysilicon film and a metal film such as aluminum, and is electrically coupled to the scan line **3a** via groove-like contact holes **42g** and **42h** passing through the gate insulating film **32** and the insulating film **42**. The semiconductor film **31a** includes a channel area **31g** overlapping with the gate electrode **33a** in plan view, a first source/drain region **31s** adjacent to the channel area **31g** on one side **Y1** in the second direction **Y**, and a second source/drain region **31d** adjacent to the channel area **31g** on the other side **Y2** in the second direction **Y**. The transistor **30** has a Lightly-Doped Drain (LDD) structure. Therefore, the second source/drain region **31d** includes a high concentration region **31d1** in which a high concentration impurity is introduced at a position away from the channel area **31g**, and a low concentration region **31d2** in which a low concentration impurity is introduced between the channel area **31g** and the high concentration region **31d1**, the concentration being lower than that in the high concentration region **31d1**. The first source/drain region **31s** includes a high concentration region **31s1** in which a high concentration impurity is introduced at a position away from the channel area **31g**, and a low concentration region **31s2** in which a low concentration impurity is introduced between the channel area **31g** and the high concentration region **31s1**, the concentration being lower than that in the high concentration region **31s1**. The semiconductor film **31a** is formed of a polysilicon film or the like, and the gate insulating film **32** has a two-layer structure including a first gate insulating film formed of a silicon oxide film obtained by thermal oxidization of the semiconductor film **31a**, and a second gate insulating film formed of a silicon oxide film formed by a low-pressure CVD method or the like.

(39) Between the gate insulating film **32** and the insulating film **43**, relay electrodes **33c** and **33d** are provided. The relay electrodes **33c** and **33d** is formed of the same conductive film as the gate electrode **33a**. A contact hole **42c** that passes through a notch **3a5** of the scan line **3a** and reaches the second conductive film **5a** is formed in the gate insulating film **32**, the insulating film **42**, and the insulating film **41**. Therefore, the relay electrode **33c** is electrically coupled to the second conductive film **5a** via the contact hole **42c**. Further, a contact hole **42d** that passes through a notch **5a5** of the second conductive film **5a** and reaches the first conductive film **4a** is formed in the gate insulating film **32**, the insulating film **42**, and the insulating film **41**. Therefore, the relay electrode **33d** is electrically coupled to the first conductive film **4a** via the contact hole **42d**.

(40) Between the insulating film **43** and the insulating film **44**, the relay electrodes **8c**, **8d**, and **8s** are formed. The relay electrodes **8c**, **8d**, and **8s** are formed of a conductive film having a light shielding property such as a metal silicide film, a metal film, or a metal compound film. The relay electrode **8d** is electrically coupled to the second source/drain region **31d** via a contact hole **43d** that passes through the insulating film **43** and the gate insulating film **32**. The end of the relay electrode **33d** is exposed at the contact hole **43d**, and hence the relay electrode **8d** is electrically

coupled to the relay electrode **33d**. The relay electrode **8s** is electrically coupled to the first source/drain region **31s** via a contact hole **43s** that passes through the insulating film **43** and the gate insulating film **32**. The relay electrode **8c** is electrically coupled to the relay electrode **33c** via a contact hole **43c** that passes through the insulating film **43**.

(41) Between the insulating film **44** and the insulating film **45**, the data line **6a** and relay electrodes **6e** and **6c** are provided. The data line **6a** and the relay electrodes **6c** and **6e** are formed of a conductive film having a light shielding property such as a metal silicide film, a metal film, or a metal compound film. The data line **6a** is electrically coupled to the relay electrode **8s** via a contact hole **44s** that passes through the insulating film **44**. The relay electrode **6e** is electrically coupled to the relay electrode **8d** via a contact hole **44e** that passes through the insulating film **44**. The relay electrode **6c** is electrically coupled to the relay electrode **8c** via a contact hole **44c** that passes through the insulating film **44**.

(42) Between the insulating film **45** and the insulating film **46**, the capacitance line **7a** and a relay electrode **7e** are provided. The capacitance line **7a** extends in the second direction Y so as to overlap with the data line **6a** in plan view. The capacitance line **7a** and the relay electrode **7e** are formed of a conductive film having a light shielding property such as a metal silicide film, a metal film, or a metal compound film. The capacitance line **7a** is electrically coupled to the relay electrode **6c** via a contact hole **45c** that passes through the insulating film **45**. The relay electrode **7e** is electrically coupled to the relay electrode **6e** via a contact hole **45e** that passes through the insulating film **45**.

(43) The pixel electrode **9a** is electrically coupled to the relay electrode **7e** via a contact hole **46e** that passes through the insulating film **46**. In the present exemplary embodiment, among four pixel electrodes **9a1**, **9a2**, **9a3**, and **9a4** adjacent to one another in the periphery of the contact hole **46e**, the pixel electrode **9a1** positioned on the other side Y2 in the second direction Y with respect to the contact hole **46e** is electrically coupled to the relay electrode **7e** via the contact hole **46e**. Therefore, the pixel electrode **9a** is electrically coupled to the second source/drain region **31d** via the relay electrode **7e**, the relay electrode **6e**, and the relay electrode **8d**. Therefore, when the transistor **30** is in an on-state, an image signal is supplied from the data line **6a** to the pixel electrode **9a**. Further, the pixel electrode **9a** is electrically coupled to the first conductive film **4a** of the capacitance element **55** via the relay electrode **7e**, the relay electrode **6e**, the relay electrode **8d**, and the relay electrode **33d**.

(44) 1-3. Configuration Example of Motherboard **100** for Image Device

(45) FIG. **6** is an explanatory view of a motherboard **100** for an image device according to the first exemplary embodiment of the present disclosure. The motherboard **100** for an image device illustrated in FIG. **6** is a large substrate for manufacturing the plurality of element substrates **10** at the same. Therefore, on the motherboard **100** for an image device, a motherboard main body **119** is formed to have substantially the same structure as the element substrate **10** including the mark M and a filling film **40**, and then is divided. Further, in a process of manufacturing the image device **1**, the electrodes, the lines, the contact holes, and the like of the element substrate **10** are formed in the following manner. In a photo-lithography process, under a state of the motherboard **100** for an image device illustrated in FIG. **6**, a resist is exposed to light via a light exposure mask, development is performed, and then a resist mask is formed. After that, in an etching process, etching is performed through an opening portion of the resist mask. Thus, before the photo-lithography process, the marks M illustrated in FIG. **6** are formed as alignment marks at the motherboard **100** for an image device, and a light exposure mask is positioned with the marks M as reference positions.

(46) The motherboard **100** for an image device includes a plurality of effective regions **110** that correspond to the element substrates **10** after division and an ineffective region **120** that is removed to obtain the element substrates **10** at the time of division. In the present exemplary embodiment, the marks M are provided on the outer side of the pixel area **10b** in each of the plurality of effective



regions **110**. Therefore, in the image device **1** illustrated in FIG. **1**, the marks **M** are present at the element substrate **10**. For example, the marks **M** are present at the positions corresponding to the four corners of the element substrate **10**.

(47) 1-4. First Configuration Example of Mark **M**

(48) FIG. **7** is a plan view illustrating the mark **M** illustrated in FIG. **1** in an enlarged manner. FIG. **8** is a cross-sectional view of the mark **M** illustrated in FIG. **7**. FIG. **9** is a plan view illustrating a mark **Mx** according to a reference example of the present disclosure in an enlarged manner. FIG. **10** is a cross-sectional view of the mark **Mx** illustrated in FIG. **9**.

(49) In FIG. **7** and FIG. **8**, the mark **M** includes a recessed portion **195** on the outer side of the pixel area **10b** of the substrate body **19**, and the filling film **40** fills the inside of the recessed portion **195**. In the present exemplary embodiment, the recessed portion **195** is formed at the same layer as the recessed portion **190** illustrated in FIG. **4** and FIG. **5**. More specifically, the recessed portion **195** is recessed in a surface of the substrate body **19** on the pixel electrode **9a** side, toward a side opposite to the pixel electrode **9a**. In the present exemplary embodiment, the substrate body **19** and the motherboard **100** for an image device are transmissive, and hence the filling film **40** includes a light-shielding film. Therefore, the mark **M** is visually recognizable when viewing the motherboard **100** for an image device from any planes.

(50) The filling film **40** is formed of the multi-layered film **550** used for forming the capacitance element **55** and the insulating film **41**, and the multi-layered film **550** includes the first conductive film **4a** and the second conductive film **5a** as light-shielding films. At the mark **M**, a light-shielding film including the first conductive film **4a** and the second conductive film **5a** is formed in a range slightly larger than the recessed portion **190**. However, a planar shape and a planar size of the light-shielding film including the first conductive film **4a** and the second conductive film **5a** is substantially the same as a planar shape and a planar size of the recessed portion **190**.

(51) Here, the recessed portion **195** includes a first groove **196** extending in the first direction **X** and a second groove **197** extending in the second direction **Y** intersecting with the first groove **196**, the second groove **197** not intersecting with the first groove **196**. In the present exemplary embodiment, the recessed portion **195** includes a plurality of second grooves **197**. More specifically, the second grooves **197** extend in the first direction **X** on both sides of a first groove **191** in the first direction **X**.

(52) Each of the plurality of second grooves **197** is prevented from intersecting with the first groove **196**, and extends in the first direction **X** intersecting with the first groove **196**. More specifically, the second grooves **197** are provided at positions away from the first groove **196** across a gap **G** in the first direction **X**, and hence do not intersect with the first groove **196**. Thus, a groove width of the recessed portion **195** is small at any position. Therefore, when the filling film **40** is formed, a recessed portion **5a6** is filled with a part accumulated on side walls of the first groove **196** and the second groove **197**, in addition to a part accumulated on bottom walls of the first groove **196** and the second groove **197**. Thus, step coverage is satisfactory. Therefore, as illustrated in FIG. **8**, for example, even in a case in which the recessed portion **5a6** generated by the first groove **196** and the second groove **197** is generated in the surface of the second conductive film **5a**, when the insulating film **41** is formed, the recessed portion **5a6** is readily filled with the insulating film **41**. Thus, in the filling film **40**, the insulating film **41** is less likely to receive a stress. With this, a crack is less likely to be generated in the insulating film **41**, and a crack is less likely to develop. Therefore, observation of the mark **M** is not hindered by a crack, and hence the mark **M** is readily observed. Further, even when chemical machine polishing is performed after formation of the insulating film **41** as in the present exemplary embodiment, a film thickness for forming the insulating film **41** can be reduced, and hence a time period for forming the insulating film **41** can be shortened.

(53) In contrast, at the mark **Mx** according to the reference example illustrated in FIG. **9** and FIG. **10**, a first groove **196x** and a second groove **197x** forming a recessed portion **195x** intersect with

each other. Thus, at a crossing portion **195y** of the first groove **196x** and the second groove **197x**, a groove width in a direction intersecting obliquely with the first groove **196x** and the second groove **197x** is wider than a design value at the time of etching. Therefore, a wide portion **195z** is present at the crossing portion **195y** of the first groove **196x** and the second groove **197x**. With this, when the filling film **40** is formed, there is a difficulty in filling the recessed portion **195x** with the filling film **40** due to the portion accumulated on bottom walls and side walls of the first groove **196x** and the second groove **197x**. For example, in a case in which a recessed portion **5a7** generated by the first groove **196x** and the second groove **197x** is generated in the surface of the second conductive film **5a**, when the insulating film **41** is formed, there is difficulty in filling the recessed portion **5a7** with the insulating film **41**. Therefore, the insulating film **41** is more likely to receive a stress. With this, a crack is more likely to be generated in the insulating film **41**, and a crack is more likely to develop. Therefore, observation of the mark M is hindered by a crack, and there is a difficulty in observing the mark M.

(54) Further, in a case in which the wide portion **195z** wider than a design value is generated at the crossing portion **195y** of the first groove **196x** and the second groove **197x**, when the multi-layered film **550** is subjected to patterning, the recessed portion **195x** projects at the crossing portion **195y** from a portion in which the multi-layered film **550** is formed. With this, there is a difficulty in observing the mark M. Further, even when chemical machine polishing is performed after formation of the insulating film **41**, the film thickness for forming the insulating film **41** needs to be increased, and hence a time period for forming the insulating film **41** is prolonged.

(55) Thus, according to the exemplary embodiment described with reference to FIG. 7 and FIG. 8, a situation where the recessed portion **195** projects from the portion in which the multi-layered film **550** is formed or the like is not caused. With this, problems that may arise at the time of formation of the mark Mx illustrated in FIG. 9 and FIG. 10 can be solved. Note that, when the substrate body **19** is a substrate having a light-shielding property such as a silicon substrate, the filling film **40** may be transmissive.

(56) 1-5. Second Configuration Example of Mark M

(57) FIG. 11 is an explanatory view illustrating a second configuration example of the mark M at the motherboard **100** for an image device according to the present disclosure. Note that, in any one of the present exemplary embodiment and exemplary embodiments described later, the basic configuration of the mark M is similar to that in the exemplary embodiment described with reference to FIG. 7 and FIG. 8. Thus, common components are denoted with the same reference signs, and description therefor is omitted.

(58) As illustrated in FIG. 11, the mark M in the present exemplary embodiment also includes the recessed portion **195**, similarly to the mark M in the exemplary embodiment described with reference to FIG. 7 and FIG. 8. In the present exemplary embodiment, the recessed portion **195** includes the plurality of first grooves **196** extending in the first direction X and the plurality of second grooves **197** extending in the second direction Y intersecting with the first grooves **196**, the second grooves **197** not intersecting with the first grooves **196**. In the present exemplary embodiment, the mark M has a cross-like shape, and the first grooves **196** and the second grooves **197** are alternately arranged along the outer edge of the mark M. Further, the first groove **196** and the second groove **197** are away from each other across the gap G, and hence the first groove **196** and the second groove **197** do not intersect with each other. Thus, the recessed portion **195** does not include the wide portion **195z** described with reference to FIG. 9 and FIG. 10. Therefore, as illustrated in FIG. 8, the recessed portion **5a6** is readily filled with the filling film **40**, and hence a crack is less likely to be generated in the insulating film **41**, for example. Thus, effects similar to those in the exemplary embodiment described with reference to FIG. 7 and FIG. 8 can be exerted. Moreover, the first grooves **196** and the second grooves **197** are arranged along the outline of the mark M, and hence widths of the first grooves **196** and the second grooves **197** are narrow. Therefore, according to the present exemplary embodiment, the recessed portion **5a6** is more

readily filled with the filling film **40** in comparison to the exemplary embodiment described with reference to FIG. 7 and FIG. 8, and hence an effect of lowering possibility of generating a crack in the insulating film **41** and other effects can be exerted.

(59) 1-6. Third Configuration Example of Mark M

(60) FIG. 12 is an explanatory view illustrating a third configuration example of the mark M at the motherboard **100** for an image device according to the present disclosure. The mark M illustrated in FIG. 12 corresponds to a modification example of the mark M illustrated in FIG. 11. In the case of the mark M illustrated in FIG. 11, the first grooves **196** and the second grooves **197** are not continuous. In the present exemplary embodiment, as illustrated in FIG. 12, the first grooves **196** and the second grooves **197** are continuous. Even in this exemplary embodiment, the first grooves **196** and the second grooves **197** do not intersect with each other. Therefore, the wide portion **195z** described with reference to FIG. 9 and FIG. 10 is less likely to be generated at the portion where the first groove **196** and the second groove **197** are continuous with each other, and hence the recessed portion **195** is readily filled with the filling film **40**.

(61) 1-7. Fourth Configuration Example of Mark M

(62) FIG. 13 is an explanatory view illustrating a fourth configuration example of the mark M at the motherboard **100** for an image device according to the present disclosure. As illustrated in FIG. 13, the mark M is an aggregate including a plurality of portions M1, M2, M3, M4, M5, and M6 that are arrayed in the first direction X. Each of the plurality of portions M1, M2, M3, M4, M5, and M6 includes the identical shape of recessed portion **195**. In the present exemplary embodiment, the recessed portion **195** includes the first grooves **196** extending in the first direction X and the second grooves **197** extending in the second direction Y intersecting with the first grooves **196**, the second grooves **197** not intersecting with the first grooves **196**. In the present exemplary embodiment, the recessed portion **195** includes the plurality of first grooves **196** and the plurality of second grooves **197**. More specifically, the mark M has a quadrangular frame-like shape, and the first grooves **196** and the second grooves **197** are alternately arranged along the outer edge of each of the plurality of portions, that are M1, M2, M3, M4, M5, and M6.

(63) Further, the first groove **196** and the second groove **197** are away from each other across the gap G, and hence the first groove **196** and the second groove **197** do not intersect with each other. Thus, the wide portion **195z** described with reference to FIG. 9 and FIG. 10 is not present. Therefore, as described with reference to FIG. 7 and FIG. 8, the recessed portion **195** is readily filled with the filling film **40**, and hence a crack is less likely to be generated in the insulating film **41**, for example. Thus, effects similar to those in the exemplary embodiment described with reference to FIG. 7 and FIG. 8 can be exerted. Moreover, the first grooves **196** and the second grooves **197** are arranged along the outline of the mark M, and hence widths of the first grooves **196** and the second grooves **197** are narrow. Therefore, according to the present exemplary embodiment, the recessed portion **195** is readily filled with the filling film **40**, and hence an effect of lowering possibility of generating a crack in the insulating film **41** and other effects can be exerted.

(64) 1-8. Fifth Configuration Example of Mark M

(65) FIG. 14 is an explanatory view illustrating a fifth configuration example of the mark M at the motherboard **100** for an image device according to the present disclosure. The mark M illustrated in FIG. 14 corresponds to a modification example of the mark M illustrated in FIG. 13. In the case of the mark M illustrated in FIG. 13, the first grooves **196** and the second grooves **197** are not continuous. In the present exemplary embodiment, as illustrated in FIG. 14, the first grooves **196** and the second grooves **197** are continuous. However, the first grooves **196** and the second grooves **197** do not intersect with each other. Therefore, the wide portion **195z** described with reference to FIG. 9 and FIG. 10 is less likely to be generated at the portion where the first groove **196** and the second groove **197** are continuous with each other, and hence the recessed portion **195** is readily filled with the filling film **40**.

(66) 1-9. Sixth Configuration Example of Mark M

(67) FIG. 15 is an explanatory view illustrating a sixth configuration example of the mark M at the motherboard 100 for an image device according to the present disclosure. In the present exemplary embodiment, the recessed portion 195 is also provided similarly to the exemplary embodiment described with reference to FIG. 7. Further, the recessed portion 195 includes the first grooves 196 extending in the first direction X and the second grooves 197 extending in the second direction Y intersecting with the first grooves 196, the second grooves 197 not intersecting with the first grooves 196. In the present exemplary embodiment, the recessed portion 195 includes the plurality of first grooves 196 and the plurality of second grooves 197. Further, the first grooves 196 and the second grooves 197 are alternately arranged. Therefore, the mark M has a quadrangular frame-like shape. Therefore, the mark M is used as an alignment mark. Further, the mark M has a frame-like shape. A photo-lithography technique is used to form a resin pattern 2a on an inner side of the mark M, and then positions of the resin pattern 2a and the mark M are inspected. With this, positional accuracy at the time of forming the resin pattern 2a by the photo-lithography technique can be evaluated.

(68) In the present exemplary embodiment, the first groove 196 and the second groove 197 are away from each other across the gap G, and hence the first groove 196 and the second groove 197 do not intersect with each other. Thus, the recessed portion 195 does not include the wide portion 195z described with reference to FIG. 9 and FIG. 10. Therefore, as described with reference to FIG. 8, the recessed portion 195 is readily filled with the filling film 40, and hence a crack is less likely to be generated in the insulating film 41, for example. Thus, effects similar to those in the exemplary embodiment described with reference to FIG. 7 and FIG. 8 can be exerted. Moreover, the first grooves 196 and the second grooves 197 are arranged along the outline of the mark M, and hence widths of the first grooves 196 and the second grooves 197 are narrow. Therefore, according to the present exemplary embodiment, the recessed portion 195 is readily filled with the filling film 40, and hence an effect of lowering possibility of generating a crack in the insulating film 41 and other effects can be exerted.

(69) 1-10. Seventh Configuration Example of Mark M

(70) FIG. 16 is an explanatory view illustrating a seventh configuration example of the mark M at the motherboard 100 for an image device according to the present disclosure. The mark M illustrated in FIG. 16 corresponds to a modification example of the mark M illustrated in FIG. 15. In the case of the mark M illustrated in FIG. 15, the first grooves 196 and the second grooves 197 are not continuous. In the present exemplary embodiment, as illustrated in FIG. 16, the first grooves 196 and the second grooves 197 are continuous. However, the first grooves 196 and the second grooves 197 do not intersect with each other. Therefore, the wide portion 195z described with reference to FIG. 9 and FIG. 10 is less likely to be generated at the portion where the first groove 196 and the second groove 197 are continuous with each other, and hence the recessed portion 195 is readily filled with the filling film 40.

(71) 1-11. Configuration Example of Capacitance Element 55

(72) FIG. 17 is a plan view illustrating a configuration example of the capacitance element 55 illustrated in FIG. 4 and FIG. 5. For forming the capacitance element 55 illustrated in FIG. 4 and FIG. 5, the mark M is formed on the outer side of the pixel area 10b at the same time when the recessed portion 190 is formed. As illustrated in FIG. 17, at the capacitance element 55, the recessed portion 190 is recessed in the region overlapping with the capacitance element 55 in plan view, toward the substrate body 19. The recessed portion 190 includes the first groove 191 extending in the first direction X and second grooves 192 extending in a direction different from the extending direction of the first groove 191. In the present exemplary embodiment, the first groove 191 linearly extends in the second direction Y so as to overlap with the semiconductor film 31a in plan view, and the plurality of second grooves 192 extend in the second direction Y on both sides of the first groove 191 in the width direction in the first direction X. Here, in the multi-

layered film **550**, the second conductive film **5a** overlaps with a side wall and a bottom wall of each of the first groove **191** and the second grooves **192** via the insulating film **49** and the first conductive film **4a**. Therefore, electrostatic capacitance of the capacitance element **55** is large. Further, depths of the first groove **191** and the second grooves **192** are deeper than the total film thickness of the multi-layered film **550**, and the multi-layered film **550** is formed uniformly by the CVD method or the like. Thus, the capacitance element **55** with stable electrostatic capacitance can be configured.

(73) Here, the second grooves **192** do not intersect with the first groove **191** extending in the second direction Y, and extend in the first direction X intersecting with the first groove **191**. More specifically, the second grooves **192** are provided at positions away from the first groove **191** across the gap G in the first direction X, and hence do not intersect with the first groove **191**. Therefore, groove widths of the first groove **191** and the second grooves **192** are narrow at any position. Therefore, as described with reference to FIG. 7 and FIG. 8, even in a case in which the recessed portion **5a6** generated by the first groove **191** and the second grooves **192** is generated in the surface of the second conductive film **5a**, when the insulating film **41** is formed, the recessed portion **5a6** is readily filled with the insulating film **41** due to the portion accumulated on bottom walls and side walls of the first groove **191** and the second grooves **192**. Therefore, even when chemical machine polishing is performed after formation of the insulating film **41** as in the present exemplary embodiment, a film thickness for forming the insulating film **41** can be reduced, and hence a time period for forming the insulating film **41** can be shortened. Further, it is only required to form the semiconductor film **31a** into a flat shape, and hence chemical machine polishing for flattening the surface of the insulating film **41** is not required as long as the recessed portion **5a6** is filled. In the present exemplary embodiment, the insulating film **41** is formed of a silicon oxide film or the like, and is formed to have a film thickness from 0.2  $\mu\text{m}$  to 0.5  $\mu\text{m}$ . Further, the insulating film **41** has a shape excellent in a step property suitable for filling the recessed portion **5a6**. Further, an opening width of the first groove **191** is from 0.6  $\mu\text{m}$  to 1.0  $\mu\text{m}$ , and a bottom wall width thereof is from 0.4  $\mu\text{m}$  to 0.8  $\mu\text{m}$ , and a depth thereof is from 1  $\mu\text{m}$  to 2  $\mu\text{m}$ .

(74) 1-12. Modification Example of Motherboard **100** for Image Device

(75) FIG. **18** is an explanatory view of a modification example of the motherboard **100** for an image device illustrated in FIG. **6**. The motherboard **100** for an image device illustrated in FIG. **6** includes the marks M provided on the outer side of the pixel area **10b** in each of the plurality of effective regions **110** that correspond to the element substrates **10** after division, while in the present exemplary embodiment, as illustrated in FIG. **18**, the marks M are provided at the ineffective region **120** that is removed after division of the element substrates **10**. Therefore, at the time when the image device **1** illustrated in FIG. **1** is configured, the mark M are not present in the image device **1**.

(76) Note that, in FIG. **18**, the marks M are arranged on the outer sides of the effective regions **110** that correspond to the plurality of element substrates **10** after division, but the marks M may be provided between regions of the effective regions **110** that correspond to the element substrates **10** after division.

## 2. Second Exemplary Embodiment: Application Example to Organic Electroluminescence Device **1b**

(77) FIG. **19** is an explanatory view of the organic electroluminescence device **1b** to which the present disclosure is applied. Note that, for describing the present exemplary embodiment and exemplary embodiments described later, an element substrate and a substrate body are described as the element substrate **10** and the substrate body **19**, respectively, so that correspondence with the liquid crystal device **1a** illustrated in FIG. **1** and FIG. **2** is easily understood.

(78) The image device **1** illustrated in FIG. **19** corresponds to the organic electroluminescence device **1b**. In the organic electroluminescence device **1b**, the substrate body **19** of the element substrate **10** is provided with light-emitting elements **115** and the like, and an image is displayed by

light emitted from the light-emitting elements **115**.

(79) In the present exemplary embodiment, at the substrate body **19**, insulating films **141**, **142**, **143**, **144**, **145**, **146**, and **147** being transmissive and forming inter-layer insulating films and the like are layered. Further, at the substrate body **19**, a semiconductor layer **131** forming a transistor **130** or the like of each of pixels **101e** is formed, and the semiconductor layer **131** is covered with a gate insulating film **132**. A gate electrode **133a** is formed between the gate insulating film **132** and the insulating film **141**. A relay electrode **133d** or the like is formed between the insulating film **141** and the insulating film **142**. A first capacitance electrode **164a** or the like is formed between the insulating film **142** and the insulating film **143**. A second capacitance electrode **165a** or the like that faces the first capacitance electrode **164a** via the insulating film **143** is formed between the insulating film **143** and the insulating film **144**. A wiring line **167a** or the like that is electrically coupled to the second capacitance electrode **165a** and has a light reflection property is formed between the insulating film **144** and the insulating film **145**. The insulating film **146** is a light path adjustment layer for setting a resonant wavelength of an optical resonator **116** to a predetermined wavelength.

(80) A pixel electrode **111** forming the pixel **101e** is formed between the insulating film **146** and the insulating film **147**. The pixel electrode **111** is formed of a light-transmissive conductive material such as ITO. The insulating film **147** overlaps with an end of the pixel electrode **111**, and defines a region of the pixel **101e**. A light emission function layer **148** is formed at the pixel electrode **111**. The light emission function layer **148** is continuously formed over the plurality of pixels **101e** in the pixel area **10b**. A common electrode **121** is formed at an entirety of the pixel area **10b**. The pixel electrode **111**, the light emission function layer **148**, and the common electrode **121** form the light-emitting element **115**. At the light-emitting element **115**, the light emission function layer **148** is configured to include a light-emitting layer formed of an organic EL material, and radiates white light as a result of the supply of current. A transport layer or an injection layer of electrons or positive holes supplied to the light-emitting layer may be provided in the light emission function layer **148**.

(81) The common electrode **121** transmits some of the light that has reached it, and also functions as a semitransparent reflection layer that reflects back the rest of the light. The radiated light from the light emission function layer **148** reciprocates between the wiring line **167a** and the common electrode **121**, and components of a particular resonant wavelength are selectively amplified. Then, the reciprocating light passes through the common electrode **121**, and is emitted to a side opposite to the substrate body **19**.

(82) A sealing body **170** is formed at the upper layer side of the common electrode **121**, over the entire area of the substrate body **19**. The sealing body **170** is formed of a multi-layered film including a first sealing layer **171**, a second sealing layer **172**, and a third sealing layer **173**. The third sealing layer **173** is an inorganic material having an insulating property such as a silicon oxide film. The first sealing layer **171** is a flattened film formed of a light-transmissive organic material such as an epoxy resin. The second sealing layer **172** is formed of silicon nitride, silicon oxide, or the like, which offers excellent water-resistant and heat-resistant properties, for example.

(83) A colored layer **181** is formed at the upper surface of the sealing body **170**. The colored layer **181** transmits light having a predetermined wavelength region. Further, a transmissive cover substrate **118** is fixed to the colored layer **181**, on the opposite side to the substrate body **19**, by an adhesive layer **117**.

(84) The organic electroluminescence device **1b** thus configured is configured for, for example, red light, green light, or blue light, and is used in a projection-type display device or the like. Further, in the organic electroluminescence device **1b**, a display device may be configured by associating each of the pixels **101e** with red light, green light, or blue light.

(85) In the organic electroluminescence device **1b** thus configured, the electrodes and the like are also formed in the photo-lithography process, similarly to the liquid crystal device **1a**. Therefore,

similarly to the liquid crystal device **1a**, the marks **M** illustrated in FIG. **6** to FIG. **16** are also formed at the effective regions **110** of the motherboard **100** for an image device in the organic electroluminescence device **1b**. Further, as illustrated in FIG. **18**, the marks **M** may be formed at the ineffective region **120** of the motherboard **100** for an image device. The mark **M** includes the recessed portion **195**, and the recessed portion **195** is filled with the filling film **40** having a light-shielding property, which is formed of the same conductive film as the gate electrode **133a**, for example. Note that, when the substrate body **19** is a substrate having a light-shielding property such as a silicon substrate and a ceramics substrate, the filling film **40** may be transmissive.

### 3. Third Exemplary Embodiment: Application Example to Mirror Device **1c**

(86) FIG. **20** is an explanatory view of the mirror device **1c** to which the present disclosure is applied. The image device **1** illustrated in FIG. **20** corresponds to the mirror device **1c**. In the mirror device **1c**, the substrate body **19** of the element substrate **10** is provided with mirrors **251**, and an image is displayed by light reflected by the mirrors **251**. In the mirror device **1c**, the plurality of mirrors **251** are arranged in a matrix shape on one surface side of the substrate body **19**, and one mirror **251** corresponds to one pixel **201e**. The mirror **251** is away from the substrate body **19**. The substrate body **19** is a silicon substrate, for example. The surface of the mirror **251** is a reflection surface **257** formed of a reflective metal film such as aluminum.

(87) The mirror device **1c** includes a first stage portion **200a** including a substrate-side bias electrode **211**, substrate-side address electrodes **212** and **213**, and the like, a second stage portion **200b** including elevated address electrodes **232** and **233** and a torsion hinge **235**, and a third stage portion **200c** including the mirror **251**. At the first stage portion **200a**, an address circuit (not illustrated) is formed at the substrate body **19**. The second stage portion **200b** includes the elevated address electrodes **232** and **233**, the torsion hinge **235**, and a mirror support portion **252**. The elevated address electrodes **232** and **233** are conducted to the substrate-side address electrodes **212** and **213** via electrode posts **222** and **223**, and are also supported by the substrate-side address electrodes **212** and **213**.

(88) A Hinge arm extends from both ends of the torsion hinge **235**. The hinge arm is conducted to the substrate-side bias electrode **211** via a hinge support portion **239**, and is also supported by the substrate-side bias electrode **211**. The mirror **251** is conducted to the torsion hinge **235** via the mirror support portion **252**, and is also supported by the torsion hinge **235**. Therefore, the mirror **251** is conducted to the substrate-side bias electrode **211** via the mirror support portion **252**, the torsion hinge **235**, the hinge arm, and the hinge support portion **239**, and a bias voltage is applied thereto from the substrate-side bias electrode **211**.

(89) In the mirror device **1c**, the torsion hinge **235** is distorted when a drive voltage is applied to the substrate-side address electrodes **212** and **213** and the elevated address electrodes **232** and **233** and the mirror **251** is inclined so as to be attracted toward the substrate-side address electrode **212** and the elevated address electrode **232** or the substrate-side address electrode **212** and the elevated address electrode **233**. Further, when application of a drive voltage to the substrate-side address electrodes **212** and **213** and the elevated address electrodes **232** and **233** is stopped and a suction force with respect to the mirror **251** is no longer present, a force of restoring the mirror **251** to a posture parallel to the substrate body **19** is exerted.

(90) Here, when the mirror **251** is inclined toward the substrate-side address electrode **212** and the elevated address electrode **232**, an on-state is established, where light emitted from a light source unit is reflected by the mirror **251** toward a projection optical system. In contrast, when the mirror **251** is inclined toward the substrate-side address electrode **213** and the elevated address electrode **233**, an off-state is established, where light emitted from the light source unit is reflected by the mirror **251** toward a light absorption device. In the off-state, light is not reflected toward the projection optical system. As a result of such drive performed in each of the plurality of mirrors **251**, light emitted from the light source unit is modulated to image light at the plurality of mirrors **251**, and is projected from the projection optical system. In this manner, an image is displayed.

(91) In the mirror device **1c** thus configured, the electrodes and the like are also formed in the photo-lithography process, similarly to the liquid crystal device **1a**. In this case, a sacrifice film is utilized to achieve a configuration in which the electrodes and the like, and the substrate body **19** are away from each other. Therefore, similarly to the liquid crystal device **1a**, the marks **M** illustrated in FIG. **6** to FIG. **16** are also formed at the effective regions **110** of the motherboard **100** for an image device in the mirror device **1c**. Further, as illustrated in FIG. **18**, the marks **M** may be formed at the ineffective region **120** of the motherboard **100** for an image device. The mark **M** includes the recessed portion **195**, and the recessed portion **195** is filled with the filling film **40** having a light-shielding property, which is formed of the same conductive film as the substrate-side address electrodes **212** and **213**, for example. Note that, when the substrate body **19** is a substrate having a light-shielding property such as a silicon substrate and a ceramics substrate, the filling film **40** may be transmissive.

#### 4. Fourth Exemplary Embodiment: Application Example to Image-Capturing Device **1d**

(92) FIG. **21** is an explanatory view of the image-capturing device **1d** to which the present disclosure is applied. The image device **1** illustrated in FIG. **21** corresponds to the image-capturing device **1d** including CMOS-type solid-state imaging elements. In the image-capturing device **1d**, in the substrate body **19** of the element substrate **10**, a plurality of photodiodes **303** as solid-state imaging elements are provided to a plurality of pixels **301e**, respectively, and an image is captured by the plurality of photodiodes **303**.

(93) In the image-capturing device **1d**, the substrate body **19** is formed of a silicon substrate. At the surface layer of the substrate body **19**, the plurality of photodiodes **303** as a plurality of photoelectric conversion portions are formed in a matrix shape in a two-dimensional manner. A charge transfer transistor for transferring a signal charge from each of the photodiodes **303** is provided adjacent to the photodiode **303** in the substrate body **19**. At a surface **319s** of the substrate body **19**, a gate insulating film **350** and a gate electrode **342** are provided. At the upper layer side of the gate electrode **342**, an insulating film **351**, a wiring line **343**, an insulating film **352**, a wiring line **344**, an inter-layer insulating film **353**, a wiring line **345**, an insulating film **354**, a wiring line **346**, an insulating film **355**, and a wiring line **347** are formed sequentially. Moreover, at the upper layer side of the wiring line **347**, an insulating film **356** and a passivation film **349** are formed sequentially. Further, the wiring lines overlapping with each other via the insulating films are electrically coupled by a contact plug (not illustrated). At the passivation film **349**, a support substrate **320** is provided via an adhesive layer **360**.

(94) A back surface **319t** of the substrate body **19**, a color filter **321** and a lens **322** are formed so as to overlap with each of the plurality of photodiodes **303**. The color filter **321** and the lens **322** are formed in the photo-lithography process. Therefore, similarly to the liquid crystal device **1a**, the marks **M** illustrated in FIG. **6** to FIG. **16** are also formed at the effective regions **110** of the motherboard **100** for an image device in the image-capturing device **1d**. Further, as illustrated in FIG. **18**, the marks **M** may be formed at the ineffective region **120** of the motherboard **100** for an image device.

(95) In the present exemplary embodiment, the mark **M** includes the recessed portion **195** passing from the passivation film **349** to the substrate body **19**. The recessed portion **195** reaches the inside of the substrate body **19**. Here, the substrate body **19** has a light-shielding property. Therefore, the filling film **410** filling the inside of the recessed portion **195** is formed of an insulating film being transmissive. In this case, the substrate body **19** is thin at the bottom portion of the recessed portion **195**, and hence is transmissive at a certain degree. Therefore, when the color filter **321** and the lens **322** are formed at the back surface **319t** of the substrate body **19**, the mark **M** can be observed from the back surface **319t** side of the substrate body **19**.

#### 5. Configuration Example of Electronic Device

(96) The liquid crystal device **1a**, the organic electroluminescence device **1b**, and the mirror device



1c are used in electronic devices such as a projection-type display device, a projection-type head up display, a direct-view-type head mounted display, a personal computer, and a digital still camera.

## Claims

1. An image device comprising: a substrate body; and a mark including: a recessed portion provided in the substrate body and a filling film filled in the recessed portion, wherein the recessed portion includes: a first groove extending along a first direction and a second groove extending along a second direction intersecting with the first direction, the second groove disconnected from the first groove in a plan view, wherein a pixel electrode for applying a voltage to an electro-optical layer, a light-emitting element, a mirror, or a photoreceptor element is provided in a pixel.
  2. The image device according to claim 1, wherein at least one of the first groove and the second groove is provided in plurality.
  3. The image device according to claim 2, wherein a plurality of the first grooves and a plurality of the second grooves are provided.
  4. The image device according to claim 3, wherein the plurality of the first grooves and the plurality of the second grooves are arranged along an outline of the mark.
  5. The image device according to claim 1, wherein the first groove and the second groove are away from each other.
  6. The image device according to claim 1, wherein the substrate body is transmissive, and the filling film includes a light-shielding film.
  7. The image device according to claim 1, wherein the substrate body has a light-shielding property, and the filling film is transmissive.
  8. A motherboard for an image device, the motherboard comprising: a motherboard main body; and a mark including: a recessed portion provided in the motherboard main body and a filling film filled in the recessed portion, wherein the recessed portion includes: a first groove extending along a first direction and a second groove extending along a second direction intersecting with the first direction, the second groove disconnected from the first groove in a plan view, wherein a pixel electrode for applying a voltage to an electro-optical layer, a light-emitting element, a mirror, or a photoreceptor element is provided in a pixel.
  9. An image device comprising: a substrate body including: a first recess provided in a pixel region, and a second recess provided outside the pixel region; a capacitance element including a multi-layered film provided in the first recess; and a mark including a filling film in the second recess including a multi-layered film identical to the multi-layered film constituting the capacitance element, wherein the first recess includes: a first groove extends along a first direction, and a second groove extends along a second direction that intersects the first direction and disconnects from the first groove in a planar view, and the second recess includes: a third groove extends along the first direction, and a fourth groove extends along the second direction and connects to the third groove in the planar view.
  10. The imaging device according to claim 9, wherein the second groove of the first recess is provided at a position away from the first groove of the first recess.
  11. The imaging device according to claim 9, wherein in the second recess, the third groove is provided to connect to one of an opposing side surfaces of the fourth groove, and the fourth groove is provided to connect to one of an opposing side surfaces of the third groove.
  12. The imaging device according to claim 9, wherein in the second recess, the third groove and the fourth groove are alternately arranged along an outer edge of the mark.
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